Benjamin Begin

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EDUCATION

University of New Hampshire (UNH), Durham, NH
Master of Science, Molecular and Cellular Biotechnology **Expected: December 2027**
Bachelor of Science, Biochemistry, Molecular and Cellular Biology **Expected: May 2027**

Relevant Coursework: Cell Culture Lab, Immunology, Molecular Genetics, Protein Biochemistry Lab, Data Science for Life Sciences, Principles of Biochemistry I & II, Principles of Cell Biology, Multidimensional Calculus

Awards:
UNH Dean's List Highest Honors **January 2024 – Present**
UNH Trustee's Scholarship – **Merit Scholarship** **August 2023 – Present**
UNH Summer Undergraduate Research Fellowship **June 2026 – August 2026**
UNH COLSA Undergraduate Research Conference – **Most Outstanding Poster Award** **April 2026**

RESEARCH EXPERIENCE

Undergraduate Research Associate, University of New Hampshire, Durham, NH **October 2025 – Present**
Advisor: Dr. Linqing Li – Dept. of Chemical Engineering and Bioengineering

Project: Engineered Biocomposites for Angiogenic Signaling and Tissue Vascularization

- Characterized how biophysical properties of Collagen I & DexMA composites influence HUVEC & HDF spreading & migration
- Identified material formulations that promote coordinated, group cell translocation, & *in situ* vasculature formation in co-culture
- Established design principles linking scaffold composition to vasculogenic outcome, laying groundwork for future paracrine signaling studies and organoid applications

PRESENTATIONS

Empower Life Sciences, New Hampshire Life Sciences, Durham, NH **September 2026**
- Selected by UNH administration to present research on DexMA-Collagen I-based biocomposite hydrogels for microvascular network formation to an audience of NH-based life science industry leaders, academic researchers, and executives.
- Represented UNH's biomaterials/tissue-engineering research at a statewide industry showcase highlighting academia-industry collaboration and career pathways in life sciences.

UNH COLSA Undergraduate Research Conference, Durham, NH **May 2026**
- Conveyed research on angiogenic biocomposite hydrogels and tissue vascularization to a variety of academic professionals as well as other UNH students at one of the largest undergraduate research conferences in the United States.

LABORATORY SKILLS

Cell Culture: Eukaryotic cell culture; Prokaryotic cell culture; Cell line management (CHO, HT1080); Centrifugation; Aspiration

Microbiology: Microbial culturing (aerobic & anaerobic); Microbe identification (Gram staining, Endospore staining, Swarm plates); Inoculation of various biochemical media; Quadrant streaking; Ames test; Catalase test; Antibiotic susceptibility testing;

Chemistry: NMR & IR spectroscopy; Column and Thin Layer Chromatography; Recrystallization; Liquid-Liquid Extraction

Biochemistry: Protein purification (e.g. affinity, gel filtration, ion exchange, size exclusion chromatography); Enzyme assays; protein quantification (Bradford assay); SDS-PAGE; Native-PAGE; Spectrophotometric analysis assays (absorbance, fluorescence);

Software: ChimeraX; ChemDraw; MNova; ImageJ (FIJI); Claude Code; Microsoft Office Suite; DaVinci Resolve; R Studio; JMP

Additional: Confocal Microscopy; Scanning Electron Microscopy; Immunofluorescence Staining; Aseptic Technique; Biosafety Level 2 Training; Hydrogel synthesis; Lyophilization; Bloodborne Pathogens; pH meter calibration; Buffer preparation; Serial dilution

PROFESSIONAL EXPERIENCE

Priority Care Ambulance Service, Canterbury, NH **May 2025 – Present**
EMT-B
- Ensured continuity of care between medical facilities, handled protected health information securely, ensured patient well-being, and navigated both familiar and unfamiliar environments efficiently.

Canobie Lake Park, Salem, NH **May 2024 – October 2024**
EMT-B
- Supervision of guest welfare, actively found, communicated (via radio), and assessed medical emergencies within the park efficiently and effectively.

LEADERSHIP AND SERVICE

UNH Men's Rugby, Durham, NH **September 2023 – Present**

President, Captain, Risk Manager

- Reinforced communication skills such as conflict resolution, problem-solving, and decision-making on and off the field with team members to communicate clearly, efficiently, and effectively.
- Managed and organized several team and community events related to philanthropy and interscholastic competition.
- Interacted and collaborated with UNH staff to coordinate team operations and secure opportunities.

Awards: *National Collegiate Rugby Scholastic All-American* **Fall 2024 – Present**

Alpha Chi Sigma (Mu Chapter), Durham, NH **September 2025 – Present**

Master of Ceremonies

- Planned and led several semi-annual initiation ceremonies for 20+ members, managing budget, venue, catering, and supply procurement in coordination with the chapter Treasurer.
- Directed a 10+ person team during the ceremony, assigning roles, running community building activities, and managing real-time event logistics.
- Delivered end-of-semester presentations and authored annual compliance reports for the national governing organization.

UNH Global Medical Brigades, Durham, NH **February 2025 – Present**

Brigade Volunteer

- Provided healthcare services to underserved communities abroad, gaining hands-on experience in public health initiatives and cross-cultural medical outreach.
- Part of the fundraising committee, established lasting relationships with local businesses to accumulate funds for future trips.

UNH Department of Molecular, Cellular, and Biomedical Sciences, Durham, NH **September 2024 – December 2024**

Learning Assistant

- Assisted students in introductory molecular and cellular biology in understanding course material.
- Moderated discussion boards to ensure students were adequately grasping and applying learned concepts.

CERTIFICATIONS

Nationally Registered Emergency Medical Technician (EMT-B) **May 2023 – Present**

- Provide basic emergency medical care and transportation for critical patients who accessed the emergency medical system.

Basic Life Support (CPR & AED) **January 2023 – Present**

- Proficient in CPR and AED use in an emergency response setting to enhance patient safety and care.